

MECHANISTIC FIXED-BED CO₂ ADSORPTION MODEL: ANALYSIS AND FORMULATION

ABSTRACT. A single component CO₂ adsorption from an inert carrier (N₂) in a 1-D fixed-bed is modelled and derived from conservation laws, closed with LDF kinetics and a nonlinear isotherm (Toth, Dual-Site Langmuir). Fitted analytical models (Yoon-Nelson, Thomas, Bohart-Adams, Clark, fractal sigmoids, etc) parameterise the shape of one curve at one operating point, with $k_{YN} = k b c_f$ and $\tau = t_{st}(u, c_f, L)$. That composition is exactly why fitted k_{YN}, τ cannot extrapolate across u, c_f, L , or T , and why the PDE model below can.

1. ASSUMPTIONS

- **A1 (1-D):** Radial gradients neglected. Caveat for the bench rig:

$$\frac{d_{\text{col}}}{d_p} \lesssim 10$$

invites wall channeling; treat radial uniformity as an idealisation to be checked, not a fact.

- **A2 (dilute feed / constant u):** CO₂ mole fraction

$$y_f \ll 1$$

so total molar flux, hence u , is z -independent. At the measured 10–15% feeds this is only first-order accurate; Remark A.1 gives the exact variable-velocity extension and the $O(y_f)$ error bound.

- **A3 (ideal gas, isobaric):** P uniform (pressure drop $\ll P$);

$$c_{\text{tot}} = \frac{P}{RT}.$$

- **A4 (Fickian axial dispersion):** All axial mixing mechanisms (molecular + Taylor–Aris + packing) are lumped into a single D_L .

- **A5 (LDF):** Intraparticle and film resistance are lumped into a single first-order driving force with constant k (§A.3); valid when one resistance dominates or concentration profiles are near-parabolic.

- **A6 (pseudo-homogeneous heat):** Gas and solid share one temperature T ; valid when inter-phase exchange is fast,

$$\frac{h_f a_p L}{u \rho_g c_{p,g}} \gg 1,$$

in which case the two-temperature model of `derivation.md` (§1.3–1.4) collapses onto this one.

- **A7 (thermodynamic consistency):** The $(-\Delta H)$ in the energy balance equals the isosteric heat implied by the isotherm’s temperature dependence (§A.4.3).

- **A8:** Constant $\varepsilon, \rho_p, c_{p,s}, \lambda_{\text{eff}}, D_L$ over the operating window.

2. GOVERNING EQUATIONS

2.1. Gas-phase CO₂ mass balance.